

TAHA TARKHANI

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Education

University of Ottawa

Expected Dec 2027

Bachelor of Applied Science in Software Engineering

Ottawa, ON

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object Oriented Programming

Experience

National Defence Canada

May 2026 – Aug 2026

Software Engineer Intern

Ottawa, ON

- Built a **Java/Spring Boot** service streaming live orbital telemetry over **WebSockets** to **200+** concurrent analysts, sustaining sub-**150ms** end-to-end latency at **500+** updates/sec.
- Accelerated a sequential satellite-tracking feed parser using a bounded **ThreadPoolExecutor** and in-memory queue, reducing **12K**-record batch ingestion from **6.5** minutes to **95** seconds.
- Implemented **Redis** caching for high-traffic ground-station API endpoints, decreasing p99 latency from **709ms** to under **100ms** while offloading repeat reads from the database.
- Shipped **React/TypeScript** features for Space-COP, a military space dashboard, including a Portal Requests card, multi-view fullscreen mode, and **3** reusable filter components to improve dashboard navigation and filtering.

National Defence Canada

Sep 2025 – Dec 2025

Software Engineer Intern

Ottawa, ON

- Optimized **PostgreSQL** queries on satellite tracking data with composite indexes and monthly table partitioning, improving peak query execution time from **1,800ms** to **210ms**.
- Automated the engineering team's deployments using a multi-stage **GitLab CI/Docker** pipeline, reducing a **25**-minute manual release process to an **8**-minute workflow.
- Increased ingest-module test coverage from **18%** to **62%** using **JUnit 5** and **Testcontainers** for PostgreSQL integration testing, catching coordinate-wraparound bugs before staging.

National Research Council Canada

Jan 2025 – Apr 2025

Data Analyst Intern

Ottawa, ON

- Improved slow analytical joins across a **PostgreSQL/MySQL** research pipeline, optimizing daily query execution from **42** to **8** minutes for **15+** researchers.
- Decreased the team's manual data-entry workload by **70%** by automating research-dataset processing via **Python** ETL scripts.
- Designed schema validation guards that quarantined malformed records, cutting weekly pipeline reruns by **88%**.

Projects

Lifts (2nd Place, MLH Hack The Hill Hackathon) | *Next.js, TypeScript, PostgreSQL/TimescaleDB, Auth0* [Live](#) | [Code](#)

- Launched a live peer-to-peer mentoring MVP mid-hackathon, driving organic adoption by **45+** teams to resolve technical blockers before the **36**-hour deadline.
- Designed an append-only **PostgreSQL/TimescaleDB** ledger that records every state change, and secured private chats for **180+** users with **Auth0** role-based access control (**RBAC**).

Request Tracer | *Java, Spring Boot, React, Docker, REST APIs*

[Live](#) | [Code](#)

- Engineered end-to-end correlation-ID propagation across **4 Spring Boot** services, ensuring Gateway, Orders, and Inventory logs share a unified trace ID in the Log Aggregator.
- Containerized the microservice architecture via **Docker Compose** and created a **React** dashboard to visualize live trace waterfall timelines, isolating a **1.6s** cold-start anomaly.

Orbit Tracker | *Python, Flask, Skyfield, globe.gl, REST APIs*

[Live](#) | [Code](#)

- Developed a **Python** service tracking **30** live satellites from CelesTrak data using **Skyfield**'s SGP4 propagation, yielding positional accuracy within **0.02%** of known GOES-16 altitudes.
- Built a pass-prediction **REST API** with an interactive globe front end (**globe.gl**), caching TLE data to accelerate repeat lookup times from **2.4s** to under **10ms**.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C++, C#, HTML/CSS

Frameworks: Spring Boot, React, Next.js, Node.js, Flask, REST APIs

Databases: PostgreSQL, MySQL, Redis, TimescaleDB, DynamoDB

Tools: Git, Docker, GitLab CI/CD, GitHub Actions, AWS (Boto3), JUnit, Testcontainers, Jest